

Aspect-Based Sentiment Analysis for the Hotel Domain (ABSA HOTEL)

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Abstract

Aspect-Based Sentiment Analysis (ABSA) plays a pivotal role in gaining deep insights into customer experiences. In this study, we focus on addressing the ABSA problem within the Vietnamese hotel domain. We present a dataset construction process utilizing the TripAdvisor platform, featuring a rigorous annotation procedure that achieved an Inter-Annotator Agreement (IAA) (Kontonatsios et al., 2023) of 0.9551 (F1-score). The study conducts experimental evaluations ranging from traditional machine learning methods to pre-trained language models. The results demonstrate that the PhoBERT model achieved superior performance with an F1-score of 73.83% on the test set, providing a solid foundation for practical applications in the tourism industry.

1 Introduction

The rapid expansion of online travel platforms in recent years has resulted in a surge in the volume of customer reviews regarding accommodation experiences. Manually analyzing this vast amount of data is infeasible, creating an urgent need to apply Sentiment Analysis techniques to automatically identify customer attitudes, thereby facilitating service quality assessment and improvement.

Traditional approaches to sentiment analysis typically assign a single sentiment label (Positive, Neutral, or Negative) to the entire review text. While this method offers the advantages of simplicity, ease of implementation, and suitability for general overview tasks, it possesses significant limitations. In practice, a customer review often addresses multiple distinct aspects of the service, where each aspect may convey contrasting emotional nuances. Relying solely on a global label fails to capture comprehensive information, resulting in the loss of granular sentiment regarding specific aspects. Consequently, hotels face difficulties

in accurately identifying which factors satisfy customers or require improvement.

To address this issue, Aspect-Based Sentiment Analysis (ABSA) is employed to accurately understand customer experiences regarding specific aspects. In this study, our objective is to apply ABSA to the hotel review domain. Specifically, our main contributions include:

- Collecting and constructing a hotel review dataset.
- Conducting sentiment analysis on specific aspects.
- Evaluating the effectiveness of various methods, including traditional Machine Learning algorithms (Logistic Regression, SVM, MultinomialNB, Random Forest) and Deep Learning models (PhoBERT).

2 Data Construction

2.1 Data Collection

The dataset was collected from TripAdvisor, a prominent online travel platform. To ensure the conciseness and focus of the feedback, we exclusively gathered reviews with a length of less than 100 words.

2.2 Annotation Schema

For each hotel review, the annotation objective is to identify two primary components: **Aspect Category** and **Sentiment Polarity**. A single review may address multiple aspects, and each aspect corresponds to a distinct sentiment. The final output is represented as a set of pairs:

$$Output = \{(Aspect, Sentiment)\}$$

Aspect Definition An aspect is defined as a combination of an **Entity** and an **Attribute**, denoted as Entity#Attribute.

The **Entities** represent the main objects within the hotel experience, including:

- **HOTEL:** The hotel in general.
- **ROOMS:** Guest rooms.
- **ROOM_AMENITIES:** In-room amenities.
- **FACILITIES:** General facilities (swimming pool, gym, etc.).
- **SERVICE:** Staff attitude and service quality.
- **LOCATION:** Hotel location.
- **FOOD&DRINKS:** Dining and beverage services.

The **Attributes** describe specific characteristics of the entity: GENERAL, PRICES, DESIGN&FEATURES, CLEANLINESS, COMFORT, QUALITY, STYLE&OPTIONS, and MISCELLANEOUS.

However, not every attribute is applicable to every entity. We constructed a constraint matrix to define valid Entity-Attribute pairs, as shown in Table 1.

Entity / Attribute	GENERAL	PRICES	DESIGN&FEATURES	CLEANLINESS	COMFORT	QUALITY	STYLE&OPTIONS	MISCELLANEOUS
HOTEL	✓	✓	✓	✓	✓	✓	×	✓
ROOMS	✓	✓	✓	✓	✓	✓	×	✓
ROOM_AMENITIES	✓	✓	✓	✓	✓	✓	×	✓
FACILITIES	✓	✓	✓	✓	✓	✓	×	✓
SERVICE	✓	×	×	×	×	×	×	×
LOCATION	✓	×	×	×	×	×	×	×
FOOD&DRINKS	×	✓	×	×	×	✓	✓	✓

Table 1: Constraint matrix of Entity-Attribute pairs valid for the annotation schema.

Sentiment Polarity Each identified aspect is assigned one of three sentiment labels:

- **Positive:** Expresses satisfaction or a compliment.
- **Negative:** Expresses dissatisfaction or a complaint.

- **Neutral:** Neutral information or unclear sentiment.

Example:

Review: “Khách sạn giá rẻ, gần biển. Nhân viên lễ tân thiếu lịch sự.” (Translate: A cheap hotel, near the beach. The reception staff were impolite.)

Labels:

- HOTEL#PRICES → Positive
- LOCATION#GENERAL → Positive
- SERVICE#GENERAL → Negative

2.3 Annotation Process

The data annotation process consists of two phases, illustrated in Figure 1 and Figure 3:

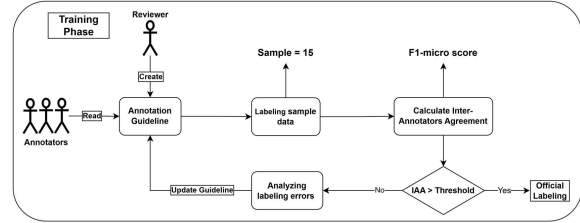


Figure 1: Annotator training process.

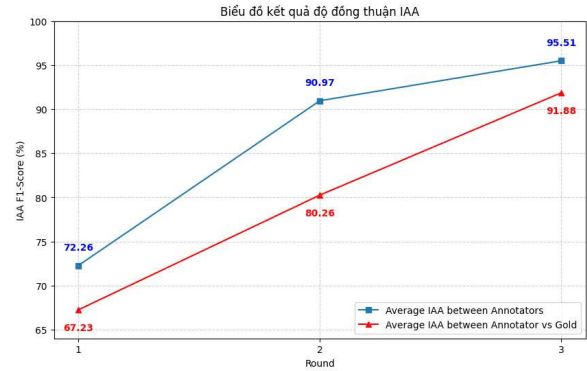


Figure 2: Inter-Annotator Agreement (IAA) results.

Annotator Training Phase: This phase involved 3 rounds to ensure accuracy and consistency among annotators. In each round, the Inter-Annotator Agreement (IAA) was calculated to assess quality. The average IAA results increased progressively: Round 1 (0.7226), Round 2 (0.9097), and peaked at Round 3 (0.9551).

Official Labeling Phase: This phase included resolving conflicting labels through discussion and voting to create the final dataset.

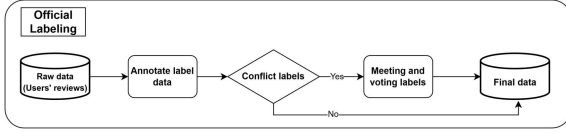


Figure 3: Official labeling process.

2.4 Data Statistics

The final dataset comprises a total of 2389 reviews, detailed in Table 2.

Dataset	No. Rev.	No. Asp.	Len.	Vocab	*
Train	1658	7109	54	5994	-
Val	359	1558	58	2558	689
Test	372	1597	56	2722	796
Full	2389	10624	55	7413	-

Table 2: Detailed data statistics. **Note:** *: number of words appearing in Test/Val but not in Train.

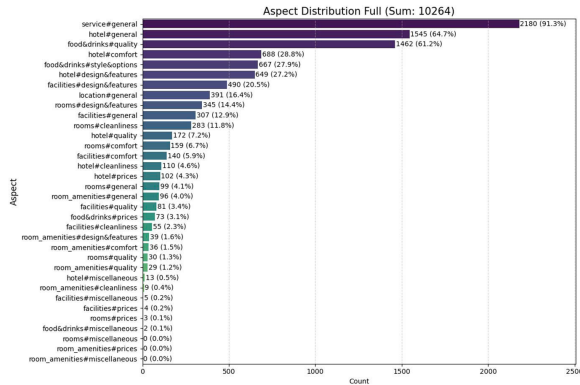


Figure 4: Distribution of aspects.

The data distribution reveals an imbalance among labels, with the most common aspects being SERVICE#GENERAL (91.3%), HOTEL#GENERAL (64.7%), and FOOD&DRINKS#QUALITY (61.2%).

Regarding sentiment, the Positive label overwhelmingly dominates over Negative and Neutral labels in the dataset.

3 Preprocessing

Review data gathered from online platforms such as TripAdvisor is typically characterized by its unstructured nature and significant noise. These reviews are often written in a free-form style, lacking strict adherence to orthographic rules, and contain various non-linguistic elements such as emojis, special characters, or internet slang (teencode). Such noise factors negatively impact the feature extraction capability and the overall performance

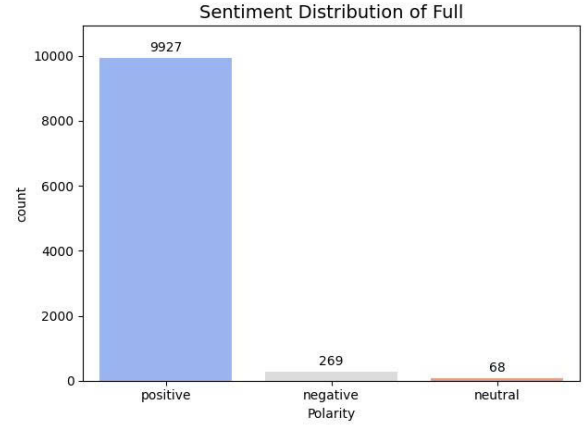


Figure 5: Distribution of sentiment.

of machine learning models. Consequently, establishing a rigorous data preprocessing pipeline is a crucial step for cleaning and normalizing the input data.

The preprocessing procedure is detailed in Figure 6.

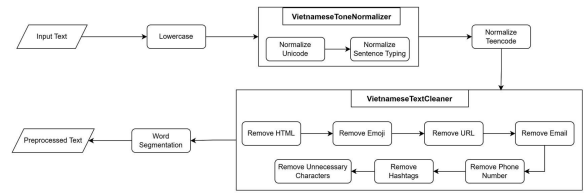


Figure 6: Data Preprocessing Workflow

Encoding and Sentence Typing Normalization. Given the specific characteristics of Vietnamese, which involves various encodings and tone placement methods, the entire dataset was converted to the standard Unicode format to prevent instances where identical words possess different binary representations. Simultaneously, variations in tone mark placement (e.g., “hòa” vs. “hoà”) were standardized to a unified form.

Intensive Text Cleaning. This phase focuses on the removal of non-linguistic elements that do not contribute value to sentiment classification, including HTML tags, emojis, special characters, hyperlinks (URLs), email addresses, and phone numbers.

Internet Slang and Abbreviation Normalization. A distinctive feature of travel reviews is the prevalence of slang, “teencode”, and inconsistent abbreviations. We utilized a mapping dictionary to convert these variants into standard language (e.g., “ko”, “k”, “kh” are normalized to “không”). This approach helps mitigate data sparsity and en-

sures that semantically identical words are processed consistently by the model.

Word Segmentation. Since Vietnamese is an isolating language, the boundaries between compound words are not clearly delimited by whitespace as they are in English. We employed RDRSegementer from the VnCoreNLP toolkit to identify Vietnamese word boundaries (e.g., "nhân viên" is converted to nhân_viên"). This step plays a decisive role in preserving the semantics of entities and attributes within the hotel aspect analysis task.

The outcome of the entire pipeline is a clean dataset with a unified lexical structure and clear segmentation, creating a solid foundation for the subsequent stages of feature extraction and machine learning model training.

4 Modeling

To address the task of Aspect-Based Sentiment Analysis (ABSA) for Vietnamese, we establish a comprehensive experimental framework comprising three primary approaches: (1) Traditional Machine Learning based on statistical features, (2) Feature-based Transfer Learning, and (3) Fine-tuning Pre-trained Language Models. An overview of these methodologies is presented in Figure 7.

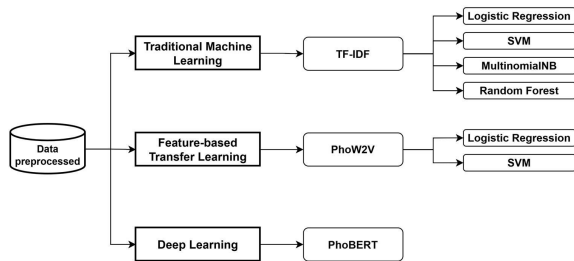


Figure 7: Overview of the modeling approaches implemented in this study.

4.1 Traditional Machine Learning

This approach serves as a baseline for evaluating the effectiveness of modern methods. Input text is represented as sparse vectors using the **TF-IDF** (Term Frequency-Inverse Document Frequency) technique. The TF-IDF value of a term t in a document d is calculated to reflect its importance within a specific context, effectively down-weighting terms that appear too frequently across the entire corpus.

Within the TF-IDF feature space, we deploy four popular classifiers:

- **Logistic Regression (LR):** A linear model that estimates the probability $P(y|x)$ via the Sigmoid function. We utilize the Liblinear optimization method, which is well-suited for sparse text problems.
- **Support Vector Machine (SVM):** This model seeks the optimal hyperplane $w^T x + b = 0$ to maximize the margin between sentiment classes (Cortes and Vapnik, 1995).
- **Multinomial Naive Bayes (MNB):** A probabilistic model based on Bayes' theorem that assumes conditional independence between features. It is often highly effective for basic text classification tasks (McCallum and Nigam, 1998).
- **Random Forest (RF):** An ensemble learning method that constructs multiple Decision Trees during training and derives the final output based on a majority voting mechanism (Breiman, 2001).

4.2 Feature-based Transfer Learning

To address the limitations regarding the sparsity and semantic deficiency of TF-IDF, we employ Word Embedding for text representation. Specifically, this study utilizes **PhoW2V** (Nguyen et al., 2020), a high-quality dense vector representation model specifically designed for Vietnamese.

PhoW2V is trained on a massive dataset of approximately 20GB (comprising Wikipedia data and Vietnamese news). In this research, we utilize the 100-dimensional vector version ($d = 100$). A word w (e.g., "nhân_viên") is mapped to a real-valued vector $\mathbb{R}^{100}$ such that semantically similar words are positioned proximally in the vector space.

The representation of an entire review is calculated as the weighted average of its component word vectors. This aggregate vector is subsequently fed into LR and SVM classifiers for sentiment prediction.

4.3 Fine-tuning Deep Learning Models

The most advanced approach employed in this study is the fine-tuning of **PhoBERT** (Nguyen and Nguyen, 2020). This is the first Pre-trained Language Model dedicated to Vietnamese, built upon the architecture of RoBERTa (Liu et al., 2019)—an optimized variant of BERT (Devlin et al., 2019).

4.3.1 PhoBERT Architecture

We utilize the **PhoBERT-base** version with the following architectural specifications:

- **Transformer Encoder Layers:** 12 layers.
- **Hidden Size:** 768 dimensions.
- **Attention Heads:** 12 heads.
- **Total Parameters:** 135 million parameters.

The model employs a multi-head Self-Attention mechanism, enabling the capture of bidirectional context for words within a sentence, thereby effectively resolving polysemy issues inherent in the Vietnamese language.

4.3.2 Fine-tuning Strategy for ABSA

To adapt PhoBERT for the multi-label ABSA task (identifying sentiment for 34 distinct aspect pairs), we designed the output architecture as depicted in Figure 8.

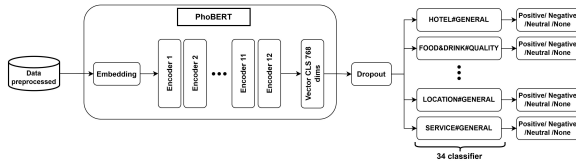


Figure 8: The PhoBERT architecture fine-tuned for the ABSA task with 34 parallel classifiers.

The processing pipeline is as follows:

1. **Input Representation:** The input review X is padded with special tokens "[CLS]" at the beginning and "[SEP]" at the end. Words are tokenized using the Byte-Pair Encoding (BPE) algorithm.
2. **Contextual Encoding:** The input sequence passes through the 12 Encoder layers of PhoBERT. The output vector corresponding to the "[CLS]" position ($h_{[CLS]} \in \mathbb{R}^{768}$) is treated as the aggregate semantic representation of the entire review.
3. **Multi-head Classification:** The vector $h_{[CLS]}$ is passed through a Dropout layer to mitigate overfitting, and is subsequently fed into 34 independent classifiers. Each classifier i (corresponding to a specific aspect, e.g., *HOTEL#GENERAL*) consists of a Linear layer combined with a Softmax function to predict the probability distribution over 4 labels: $\{Positive, Negative, Neutral, None\}$.

The loss function is calculated as the sum of the Cross-Entropy Losses from all 34 classifiers, and the model is optimized end-to-end via backpropagation.

5 Experiments & Results

5.1 Experimental Setup

To ensure objectivity and maximize the performance of traditional Machine Learning models (SVM, Logistic Regression, Random Forest, MultinomialNB), we implemented an automated hyperparameter tuning process. Specifically, the **Optuna** framework (Akiba et al., 2019) was utilized with the Tree-structured Parzen Estimator (TPE) algorithm to search for the optimal hyperparameter space.

For the Deep Learning model (PhoBERT), we employed the "phobert-base" architecture with a batch size of 16 and a learning rate of $2e^{-5}$. The model was trained for 30 epochs using a checkpointing strategy based on the F1-score on the Validation set.

The primary evaluation metric employed is the **Weighted F1-score**, which is well-suited for multi-label classification tasks exhibiting class imbalance among sentiment categories.

5.2 Results

Table 3 presents the detailed results of the experimental methods.

Approach	Feature	Model	F1-score	
			Val	Test
Machine Learning	TF-IDF	Logistic Regression	68.82	70.33
		Linear SVC	68.27	70.16
		Non-Linear SVC	64.47	65.95
		Multinomial NB	63.03	63.08
		Random Forest	67.23	68.78
	PhoW2V	Logistic Regression	63.63	65.25
		Linear SVC	64.11	64.32
		Non-Linear SVC	63.98	64.36
Deep Learning	Last hidden state	PhoBERT	72.29	73.83

Table 3: Experimental results on the Validation and Test sets. The highest values are highlighted in bold.

5.3 Analysis & Discussion

Based on the experimental results, we derive three key observations:

First, the superiority of Pre-trained Language Models.

PhoBERT achieved the highest performance with an F1-score of 73.83% on the Test set, significantly outperforming traditional methods. This confirms the efficacy of the Transformer architecture and the Self-Attention mechanism in capturing bidirectional context. Unlike ML models that rely solely on word frequency or static word vectors, PhoBERT comprehends word polysemy within diverse Vietnamese contexts, thereby better addressing complex reviews.

Second, the effectiveness of TF-IDF compared to Word Embeddings.

A notable finding is that models utilizing traditional TF-IDF features (achieving 70%) outperformed those using PhoW2V Word Embeddings (achieving 65%). We attribute this to the following reasons:

- Hotel reviews are typically short and contain keywords with high sentiment distinctiveness (e.g., "tê", "tuyệt vời", "sạch sẽ"). TF-IDF is highly effective in assigning high weights to these keywords.
- PhoW2V provides static embeddings; averaging vectors for an entire sentence may smooth out critical sentiment signals, leading to information loss in short sentences.

Third, the performance of linear classifiers.

Within the Machine Learning group, *Logistic Regression* and *Linear SVC* proved to be the most stable and effective algorithms (both exceeding 70% F1-score with TF-IDF). This suggests that with sparse, high-dimensional text data such as TF-IDF, linear separation models are less prone to overfitting and generalize better than complex non-linear models like Non-Linear SVC or Random Forest in this data context.

6 Conclusion

The research successfully constructed a process and a standard labeled dataset for Aspect-Based Sentiment Analysis (ABSA) in the Vietnamese hotel domain with high inter-annotator agreement (IAA 0.9551). Experimental results confirm that the PhoBERT model achieved superior performance (F1-score 73.83%), demonstrating a significantly better capability to capture context compared to traditional machine learning methods. Notably, given the specific nature of short reviews, the study indicates that the TF-IDF technique remains more effective than static Word Embeddings (PhoW2V).

Future development directions will focus on addressing data imbalance and optimizing the model for real-world applications.

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